An Expert-Reviewed NDVI–Soil Index Framework Using Multispectral UAV Imagery for Land Degradation Classification: Supporting Agripreneurship and Sustainable Agricultural Decision-Making

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Abstract. High-resolution multispectral unmanned aerial vehicle (UAV) imagery can support field-scale land-degradation screening and resource prioritization, but dense independent labels are often unavailable. This study develops an expert-reviewed NDVI–Soil Index framework that combines rule-derived pseudo-labels, U-Net probability maps, and XGBoost tile classification. The verified archive contains 437 DJI M3M scenes acquired over agricultural land in Karawang, Indonesia, on 2 May 2026 from 16:17:03 to 16:31:53 western Indonesian time. Every scene includes RGB, green, red, red-edge, and near-infrared files and an RTK geotag; the camera-position convex hull covers 1.139 ha. Normalized Difference Vegetation Index and a red–green Soil Index generated four ordered spectral surface-condition classes used as land-degradation proxies. A scene-level 70:15:15 split produced 81,130 training, 17,556 validation, and 17,556 test tiles. U-Net achieved 0.9519 pixel accuracy, 0.8560 intersection over union, and 0.9194 Dice score against the pseudo-label masks. An XGBoost baseline using 14 spectral and texture features achieved 0.9777 tile accuracy and 0.9690 macro F1. Adding ten U-Net probability features increased accuracy to 0.9929 and macro F1 to 0.9858. McNemar, Wilcoxon, and paired t tests favored the cascade on the paired test tiles ($p < 10^{-40}$ for the latter two tests). An agronomy-qualified expert reviewed the framework, and the authors retain signed evidence of that review. The resulting maps can support agripreneurs in prioritizing scouting, input planning, and rehabilitation monitoring, but the reported metrics measure reproduction of rule-derived targets rather than independent field-ground-truth accuracy.

Keywords: multispectral UAV, NDVI, Soil Index, expert review, agripreneurship, sustainable agriculture, decision support.

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1 Introduction

Field-scale information on vegetation condition and exposed soil is useful for agricultural monitoring, rehabilitation planning, and targeted inspection. Conventional surveys provide direct evidence but are time-consuming to repeat over spatially heterogeneous fields. UAV remote sensing offers flexible acquisition and centimetre-scale spatial detail, while multispectral sensors extend RGB observations with red-edge and near-infrared responses that are useful for vegetation and soil analysis.[1–3]

For agripreneurs, producer groups, and agricultural service providers, timely screening can inform where limited labour, field visits, and management resources should be directed. Sustainable decision-making requires more than maximizing classification performance: the evidence chain, uncertainty, and limits of each spectral category must remain visible so that interventions are not prescribed solely from an automated map. A practical framework should therefore couple reproducible remote-sensing outputs with expert review and field confirmation at consequential decision points.

Vegetation indices compress spectral contrast into interpretable indicators. The Normalized Difference Vegetation Index (NDVI) is commonly associated with vegetation density and vigour, whereas lower values can indicate sparse cover, stress, water, shadow, or exposed soil. A complementary soil-response index can help distinguish changes that are not represented by vegetation contrast alone.[4,5] Nevertheless, converting continuous spectral values into discrete condition classes requires reference data and context-specific calibration.

Dense pixel annotations are rarely available for local UAV projects. Weak supervision addresses this limitation by producing provisional labels from rules, incomplete annotations, or other indirect sources.[6] Such pseudo-labels can bootstrap segmentation, but evaluation against the same rule-derived targets measures reproduction of the labeling rule rather than independent biophysical accuracy. This distinction is particularly important when predictors and targets share spectral information.

U-Net preserves fine spatial structure through encoder–decoder skip connections.[7,8] XGBoost provides a complementary tile-level model that can combine summary statistics, histograms, texture, and learned probability features.[9,10] A cascade can therefore use U-Net not only as a final segmenter but also as a feature generator for a compact gradient-boosted classifier.

This study makes four contributions. First, it provides a verified 437-row crosswalk from scene identifiers to original filenames, acquisition time, RTK position, and altitude. Second, it implements a transparent U-Net–XGBoost cascade in which an ablation separates 14 handcrafted spectral/texture features from ten learned U-Net probability features. Third, it reports paired statistical tests and explicitly limits the interpretation to pseudo-label agreement. Fourth, it incorporates expert review of the NDVI–Soil Index framework and defines a human-in-the-loop decision layer that translates spectral screening into prioritized inspection categories without representing them as diagnoses or autonomous prescriptions. The study asks whether U-Net probability summaries improve tile-level reproduction of ordered NDVI–Soil Index pseudo-classes and how those outputs can responsibly support agripreneurial and sustainable agricultural decisions while field-valid accuracy remains to be established.

2 Materials and Methods

2.1 Study area and verified acquisition provenance

The dataset covers agricultural fields adjacent to Jl. Siliwangi in Karawang, West Java, Indonesia. The source directory `DJI_202605021610_001_Create-Area-Route2` contains 437 sequential scene folders. Each scene contains one RGB JPEG and four 16-bit GeoTIFF bands: green (G), red (R), red edge (RE), and near infrared (NIR). The resulting inventory is 437 RGB files plus 1,748 single-band files, or 2,185 files in total.

Embedded DJI XMP metadata were parsed from every RGB file. All 437 scenes contain RTK geotags (RTK flag 16), local timestamps, and altitude values. Acquisition began at 16:17:03 and ended at 16:31:53 western Indonesian time on 2 May 2026, for an elapsed duration of 890

s. Relative altitude ranged from 49.839 to 50.144 m. Latitude ranged from $-6.296305055°$ to $-6.295121261°$ and longitude from $107.308345253°$ to $107.309578910°$.

Figure 1 overlays the ordered camera path and the convex hull of all camera positions on satellite imagery. The hull area is 1.139 ha and the ordered camera-centre path is 1,913.5 m. This is acquisition geometry, not a cadastral parcel boundary or an image-footprint union. An author-supplied reference coordinate at $-6.2954927°$, $107.3085972°$ lies beside the western road boundary. A second supplied point was about 0.68 km away and is not part of this verified flight footprint; it was therefore excluded from the mapped study polygon. No unsupported 2-ha-per-side geometry is used in the analysis.



Fig 1 Verified acquisition geometry from the 437 DJI M3M RTK camera positions. Cyan lines follow the time-ordered camera centres, the red boundary is their convex hull, and the white star marks the author-supplied Location 2 reference point. The 1.139-ha camera-position hull is not a cadastral boundary or a field-label polygon. Satellite imagery: Google, Airbus, and Maxar Technologies, accessed 10 September 2026.

Figure 2 illustrates the paired RGB and multispectral observations retained in the processed archive. The complete supplementary manifest maps each `Scene_0001`–`Scene_0437` identifier to its five exact source filenames, local acquisition time, coordinates, relative and absolute altitude, RTK flag, and file sizes.

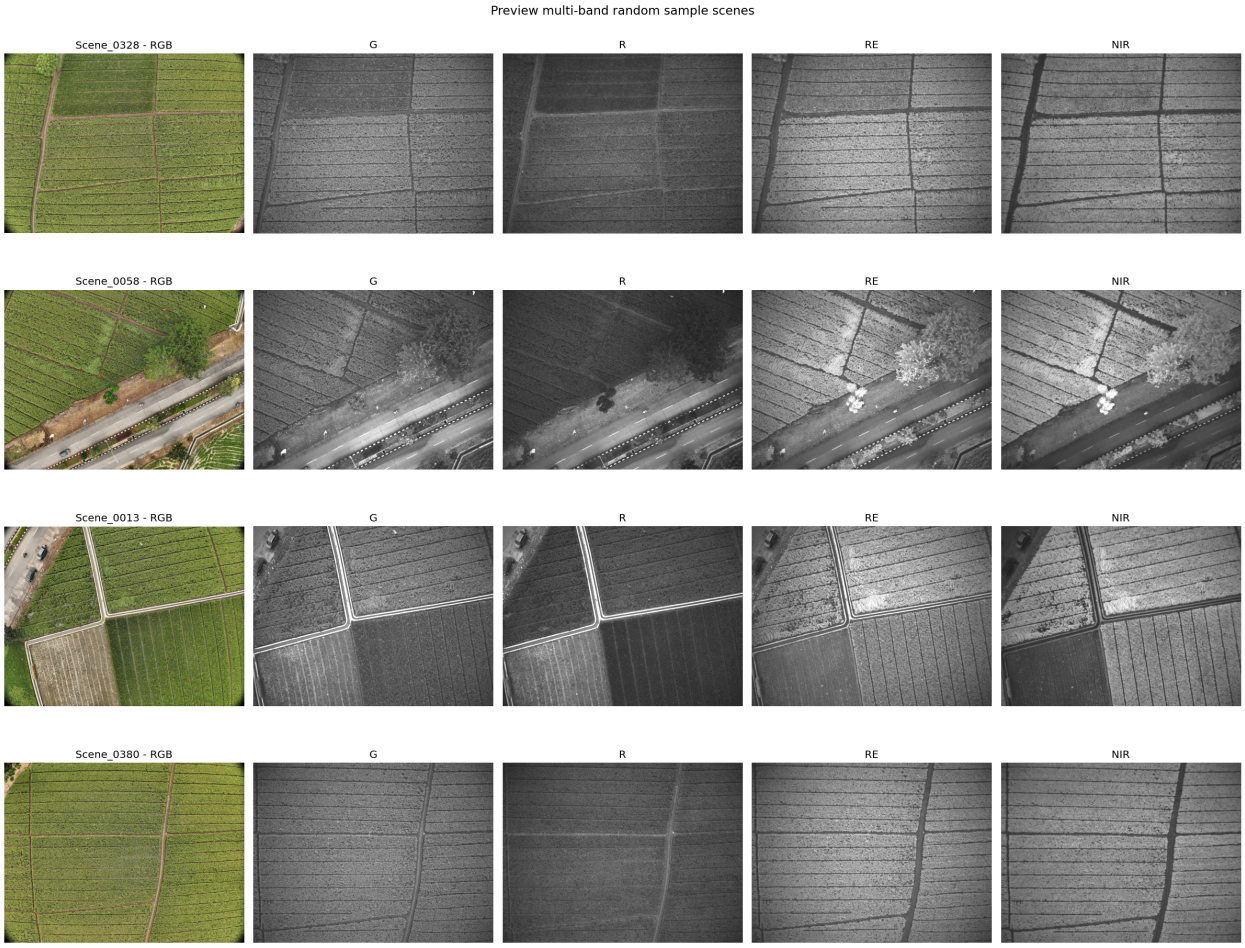


Fig 2 Representative RGB, green, red, red-edge, and near-infrared observations from four scenes in the verified 2026 archive.

2.2 Preprocessing and spectral pseudo-labels

Each unsigned 16-bit digital number was normalized to the interval from 0 to 1:

$$B_{\text{norm}} = \frac{B_{\text{raw}}}{65{,}535}. \tag{1}$$

This operation is numeric normalization and should not be interpreted as reflectance calibration.[11] NDVI and the red–green Soil Index (SI) were then calculated as

$$\text{NDVI} = \frac{NIR - R}{NIR + R + \varepsilon}, \tag{2}$$

$$\text{SI} = \frac{R - G}{R + G + \varepsilon}, \tag{3}$$

where ε prevents division by zero. NDRE, GNDVI, a modified BSI, a colour index, and a brightness index were calculated for exploratory summaries, but the target masks were generated only from NDVI and SI.[12–14]

Four ordered pseudo-classes were assigned: healthy (0), slight (1), moderate (2), and severe (3). The operational thresholds were NDVI 0.50, 0.35, and 0.20 and SI 0.10, 0.20, and 0.35:

$$\text{Healthy: NDVI} \geq 0.50 \ \wedge \ \text{SI} < 0.10, \tag{4}$$

$$\text{Slight: } 0.35 \leq \text{NDVI} < 0.50 \ \vee \ 0.10 \leq \text{SI} < 0.20, \tag{5}$$

$$\text{Moderate: } 0.20 \leq \text{NDVI} < 0.35 \ \vee \ 0.20 \leq \text{SI} < 0.35, \tag{6}$$

$$\text{Severe: NDVI} < 0.20 \ \vee \ \text{SI} \geq 0.35. \tag{7}$$

Rules were applied from severe to healthy so that a severe indicator was not overwritten by a milder class. These cutoffs are reproducible operational rules, not universal agronomic boundaries. The terms *healthy*, *slight*, *moderate*, and *severe* denote ordinal spectral surface-condition categories within this workflow.

2.3 Scene split and tile extraction

Scene identifiers were sorted and divided reproducibly with random seed 42 into 305 training scenes, 66 validation scenes, and 66 test scenes (70:15:15). Splitting before tile extraction prevents overlapping tiles from the same scene appearing in different partitions. Each scene was divided into 256×256-pixel tiles with stride 128. This produced 81,130 training tiles, 17,556 validation tiles, and 17,556 test tiles. Training-only augmentation included horizontal and vertical flips, random 90° rotation, and brightness–contrast adjustment.[15,16]

2.4 U-Net probability segmentation

The segmentation network used a U-Net with a ResNet34 encoder, ImageNet initialization, two input channels (NDVI and SI), and four output logits. The first convolution was adapted to the two-channel input. Cross-entropy and multiclass Dice losses were added:

$$L_{\text{total}} = L_{\text{CE}} + L_{\text{Dice}}. \tag{8}$$

The model was optimized with Adam for ten epochs, a learning rate of 10^{-4}, batch size 8 on an NVIDIA Tesla T4 GPU, and the validation intersection over union (IoU) as the checkpoint criterion. Softmax transformed logits into four per-pixel probability maps. The best checkpoint occurred at epoch 9.

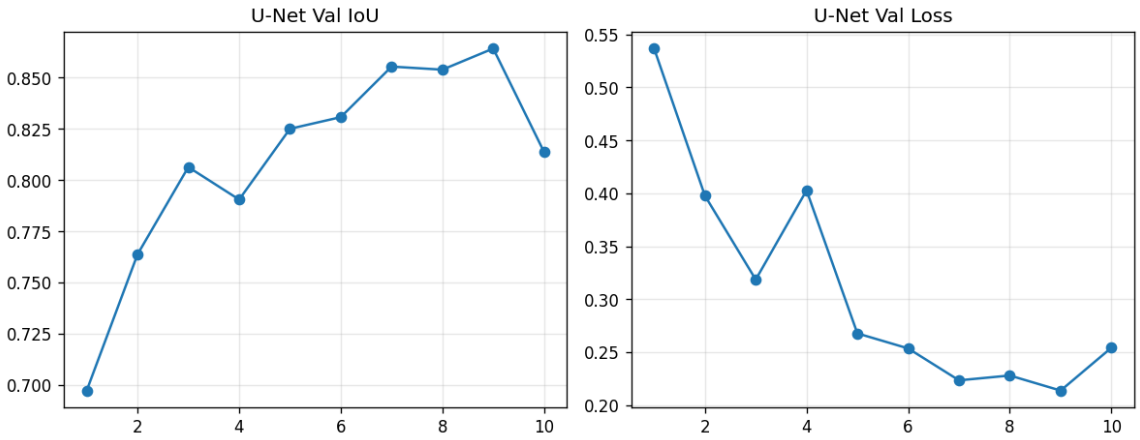


Fig 3 U-Net validation IoU and validation loss over ten epochs. The highest validation IoU, 0.8642, occurred at epoch 9.

2.5 XGBoost baseline and probability cascade

The tile target was the dominant pseudo-class in its mask. The 14-feature baseline comprised NDVI mean and standard deviation, eight NDVI histogram bins, SI mean and standard deviation, and gray-level co-occurrence-matrix contrast and homogeneity. The cascade appended ten statistics from the U-Net probability maps: class-wise means and standard deviations (eight features), mean predictive entropy, and mean dominant-class confidence. XGBoost was trained once on the 14-feature baseline and once on the full 24-feature cascade.[9,17]

This design isolates the contribution of learned probability features while preserving the same scenes, tile targets, and evaluation partitions. Figure 4 summarizes the executed pipeline.

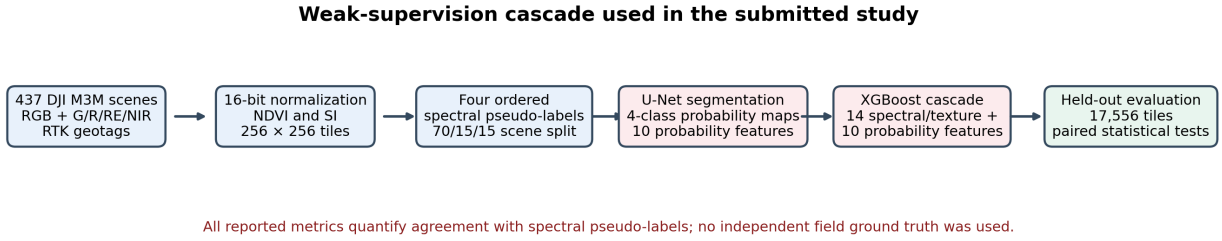


Fig 4 Executed weak-supervision cascade. U-Net probability summaries extend the 14 handcrafted tile descriptors to form the 24-feature XGBoost input.

2.6 Evaluation and statistical analysis

U-Net was evaluated at pixel level using overall accuracy, class-averaged IoU, Dice/F1, precision, and recall. XGBoost was evaluated at tile level using overall accuracy and macro-averaged precision, recall, and F1. Pixel and tile accuracies therefore answer different questions and are not treated as a direct architecture ranking.

The two XGBoost variants generated paired correctness indicators for the same 17,556 test tiles. McNemar's test examined discordant predictions. A one-sided Wilcoxon signed-rank test and a one-sided paired t test evaluated whether cascade correctness exceeded baseline correctness. The significance level was 0.05.

The target masks were derived from the same NDVI–SI information available to the models. All metrics consequently quantify agreement with the rule-derived pseudo-labels. They do not constitute field accuracy, crop-health diagnosis, soil-quality measurement, or criterion validity against independently observed plots.

2.7 Expert review and decision layer

The framework separates automated spectral screening from expert adjudication. An agronomy-qualified expert reviewed the NDVI–Soil Index framework, and signed evidence of that review is retained by the authors. For each tile, the operational review record contains the dominant pseudo-class, class probabilities, predictive entropy, RTK-linked scene identifier, acquisition time, and a recommended review priority. Healthy-category outputs receive routine monitoring priority; slight-category outputs prompt targeted visual inspection; moderate-category outputs receive priority scouting; and severe-category outputs are escalated for urgent expert and, where feasible, field assessment. These responses are workflow priorities rather than treatment recommendations.

The completed expert review supports the conceptual and operational framing of the framework; it is not represented as independent scene- or plot-level ground truth. To conduct a separate accuracy-validation study, the reviewing expert should assign an independent reference class without viewing the model prediction, record the evidence used, delineate or confirm the associated plot polygon, and sign and date each decision. A blank georeferenced adjudication template and protocol are supplied with the supplementary materials. Because independently assigned plot labels were not available for this study, no expert-agreement or field-accuracy statistic is reported.

3 Results

3.1 Dataset integrity and flight geometry

The provenance audit found 437 complete five-file scenes without a missing RGB or multispectral band. Every RGB file yielded a valid RTK location and local timestamp. The sequential filename timestamps agree with the parsed XMP acquisition interval. The first scene is `DJI_20260502161703_0001_D.JPG` at $-6.295830947°$, $107.308401472°$; the last is `DJI_20260502163153_0437_D.JPG` at $-6.295527092°$, $107.309529902°$. The lawnmower pattern in Fig. 1 is spatially coherent with an automated area route over one field block on one side of the road.

The verified camera-position hull is 1.139 ha. Because camera centres do not represent image edges, this number is not the orthomosaic footprint. It nonetheless provides a reproducible, conservative acquisition geometry and replaces the earlier unsupported assumption of 2 ha on each side of the road.

3.2 Held-out performance

Table 1 reports the executed notebook outputs. U-Net attained 0.9519 pixel accuracy, 0.8560 IoU, and 0.9194 Dice/F1. At tile level, XGBoost with 14 handcrafted features attained 0.9777 accuracy and 0.9690 macro F1. The 24-feature cascade increased these values to 0.9929 and 0.9858, respectively.

Table 1 Held-out agreement with NDVI–SI pseudo-labels

Model	Evaluation unit	Accuracy	IoU	Dice/macro F1
U-Net	Pixel	0.9519	0.8560	0.9194
XGBoost baseline, 14 features	Tile	0.9777	–	0.9690
XGBoost cascade, 24 features	Tile	0.9929	–	0.9858

The cascade improvement over the baseline was 1.52 percentage points in tile accuracy and 1.69 percentage points in macro F1. Of the paired test tiles, 332 were corrected only by the cascade and 66 were correct only under the baseline. McNemar's statistic was 176.445. The Wilcoxon statistic was 66,234 with $p = 7.40 \times 10^{-41}$, and the paired t statistic was 13.401 with $p = 4.73 \times 10^{-41}$. The reported effect on the binary correctness scale was 0.01515.

3.3 Class-level behavior and feature contribution

Table 2 shows class-level F1. The cascade improved the slight, moderate, and severe categories relative to the baseline. Its lowest F1 was 0.9756 for the least frequent healthy category (208 test tiles), while severe had F1 0.9968 with 11,884 tiles. This support distribution indicates substantial class imbalance and makes macro F1 essential alongside overall accuracy.

Table 2 Class-level F1 against the rule-derived test targets

Model	Healthy	Slight	Moderate	Severe
U-Net (pixel)	0.9190	0.9337	0.9317	0.9756
XGBoost baseline (tile)	0.9713	0.9721	0.9443	0.9881
XGBoost cascade (tile)	0.9756	0.9878	0.9832	0.9968

U-Net probability summaries dominated the cascade's gain importance. Mean severe probability ranked first (0.2807), followed by mean slight probability (0.1522) and mean NDVI (0.1436). Mean moderate probability, probability standard deviations, and selected NDVI histogram bins supplied most of the remaining importance. Figure 5 combines overall agreement, class-level F1, and the ten most important cascade features.

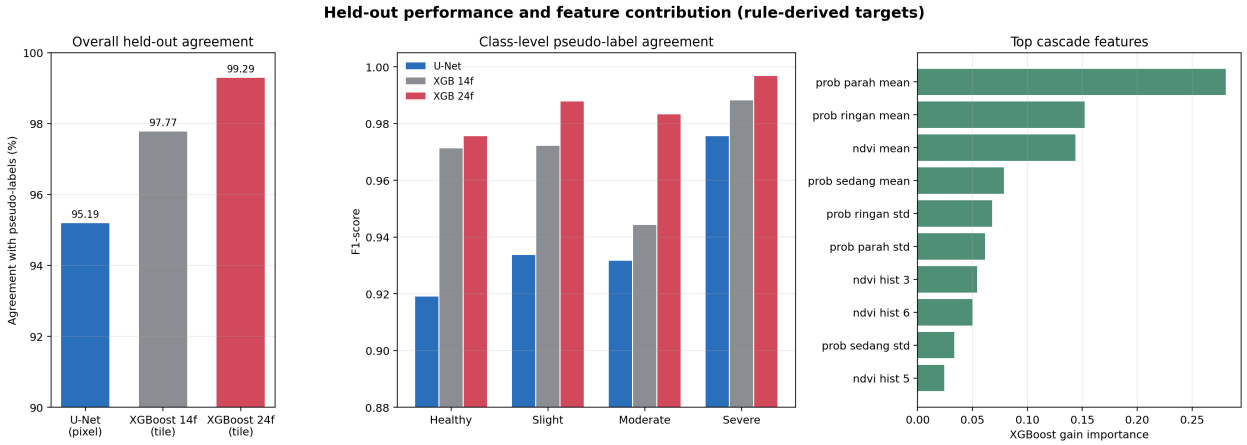


Fig 5 Held-out pseudo-label agreement and XGBoost feature contribution. U-Net values are pixel-level; XGBoost values are tile-level.

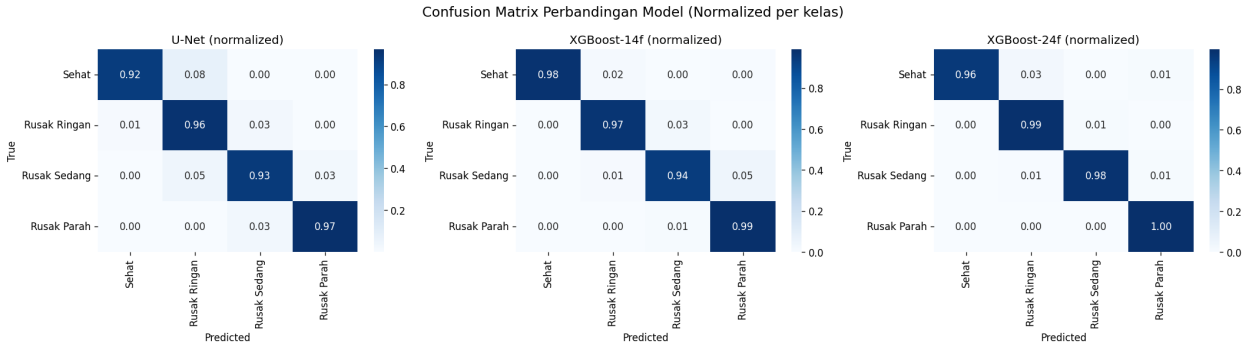


Fig 6 Normalized confusion matrices from the executed notebook for U-Net, the 14-feature XGBoost baseline, and the 24-feature cascade. Labels in the source output translate as healthy, slight, moderate, and severe.

4 Discussion

4.1 Contribution of the cascade

The ablation supports the central hypothesis: summaries of the U-Net probability field add information beyond tile-level NDVI, SI, histograms, and texture. The probability means encode the model's spatially learned evidence for each class, while entropy and dominant confidence summarize uncertainty. XGBoost can then combine these learned descriptors with explicit spectral features. The large number of baseline errors corrected by the cascade and the paired tests show that the increase is systematic within the held-out pseudo-label task.

The best validation checkpoint preceded the final training epoch. Validation IoU increased from 0.6973 at epoch 1 to 0.8642 at epoch 9 and then declined to 0.8137 at epoch 10. Retaining the best checkpoint therefore prevented the final-epoch deterioration from determining test performance. The remaining confusion was concentrated between adjacent ordinal categories, consistent with threshold transitions rather than arbitrary class swaps.

4.2 Geospatial reproducibility

The complete 437-scene crosswalk materially improves reproducibility. Every analyzed scene can now be linked to an exact RGB filename, four band filenames, timestamp, RTK coordinate, altitude, and file size. The ordered route and camera-position hull are supplied as GeoJSON, KML, and KMZ files. These records establish where and when the imagery was acquired and correct the earlier uncertainty caused by using two isolated map pins.

They do not, however, establish the true condition class of any field plot. A flight polygon answers *where the sensor was*; ground truth answers *what the surface condition actually was*. The two evidence types are complementary but not interchangeable.

4.3 Limitations and appropriate interpretation

The principal limitation is target circularity. NDVI and SI generate the pseudo-labels, and the same variables are included in the model inputs or handcrafted features. High agreement is therefore expected when the models reproduce threshold boundaries successfully. The cascade accuracy of 99.29% must not be described as field-validated land-condition accuracy.

Additional limitations are the single site and acquisition time, digital-number normalization without a documented reflectance-panel workflow, class imbalance, and the absence of independently delineated plot polygons or contemporaneous field measurements. An agronomy-qualified expert reviewed the framework, with signed evidence retained by the authors; however, that evidence was not a completed georeferenced scene-level adjudication dataset. Consequently, this paper reports neither an expert-agreement percentage nor a claim that expert review converted the masks into ground truth.

A defensible independent validation would use georeferenced plots assessed at or near acquisition time. Each plot should receive a class based on recorded crop cover, visible stress or damage symptoms, exposed-soil fraction, and, when relevant, soil or plant measurements. Those labels must be created without viewing the model prediction. Agreement between independently labeled polygons and model outputs could then be quantified with a plot-level confusion matrix, macro F1, balanced accuracy, and confidence intervals. Historical expert interpretation of the archived imagery can add content validity, but it should remain separate from field ground truth.

4.4 Operational implications

Within its stated scope, the workflow can prioritize manual review. The XGBoost cascade provides rapid dominant-class screening, while the U-Net probability map preserves within-tile spatial vari-

ation and uncertainty. Tiles near rule thresholds or with high entropy should be prioritized for expert adjudication. This human-in-the-loop use is safer than treating the pseudo-class map as a final agronomic diagnosis.

For agripreneurial operations, the framework offers a traceable basis for scheduling field scouting, documenting land-condition trends for farm-service clients, targeting follow-up measurements, and comparing rehabilitation areas over time. These functions can reduce indiscriminate field visits and focus limited inputs on areas requiring confirmation. They may support more sustainable allocation of labour, water, fertilizer, and rehabilitation resources, but no input-rate, yield, profitability, credit, insurance, or land-valuation decision should be made from the pseudo-class alone. Those decisions require agronomic context, costs, weather and crop-stage information, and expert or field verification.

5 Conclusion

This study implemented an expert-reviewed NDVI–Soil Index framework for multispectral UAV land-degradation screening. A verified archive of 437 DJI M3M scenes was linked to exact source filenames and RTK geotags from a 2 May 2026 flight. The actual camera-position hull covered 1.139 ha on one side of Jl. Siliwangi, resolving the earlier ambiguous map reconstruction.

U-Net achieved 95.19% pixel agreement with NDVI–SI pseudo-label masks. At tile level, adding ten U-Net probability features to a 14-feature XGBoost baseline raised accuracy from 97.77% to 99.29% and macro F1 from 0.9690 to 0.9858, with paired tests favoring the cascade. These values demonstrate accurate reproduction of rule-derived targets and support label bootstrapping and review prioritization. The expert-reviewed framework can assist agripreneurs and agricultural service providers by directing attention to locations that merit inspection and by doc-

umenting follow-up priorities, while keeping consequential decisions under human oversight. It does not establish independent field accuracy. Future work should obtain blinded plot-level expert labels, create georeferenced reference polygons, and evaluate the resulting model against observations that are independent of NDVI and SI.

Disclosures

The authors declare that there are no financial interests, commercial affiliations, or other potential conflicts of interest that could have influenced the objectivity of this research or the writing of this paper.

Code and Data Availability

The submitted supplementary package contains the 437-row scene-provenance manifest, acquisition summary, RTK route and hull in GeoJSON/KML/KMZ, notebook-generated evaluation tables, and tile-level predictions. The source imagery and trained model files are available from the corresponding author subject to ownership and transfer permissions. The processing notebook is available at https://colab.research.google.com/drive/1B4Lm8_WCG1JxlMW-uh-EvHvtTf0oXk7C. The notebook and supplementary records should be archived in a persistent repository if the manuscript is accepted.

Acknowledgments

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Author Contributions

Conceptualization, K.A.B. and E.S.; methodology, K.A.B. and C.D.; software and data curation, K.A.B.; formal analysis and visualization, K.A.B.; supervision, E.S., C.D., and I.R.W.; writing—original draft, K.A.B.; writing—review and editing, all authors.

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